

1. **Player Movement & Rotation:** W/S move forward/back; A/D turn (yaw). Blocked by walls, crates, trees, closed gates, holes, and fragile/broken tiles.
2. **Crate Pushing (Grid-Continuous):** walking into a crate pushes it along the nearest cardinal axis. Crates snap tile-by-tile if the next tile is free (no walls, crates, holes, trees, or closed gates).
3. **Pressure Plates:** short cylinders on the floor. Inactive = amber, Active = green when a crate sits on them. Each plate gives +30 score once.
4. **Linked Gates & Walls:** gates only open when all plates are active (or within a short “grace” window). Closed = red, open = green. Opening gives +100 score once; walls disable when gates are open.
5. **Exit Portal:** invisible sphere at the open gate mouth. Reaching it clears the level (+200 score, plus a time bonus). If gates are closed, the portal blocks entry.
6. **HUD Overlay:** on-screen text shows Lives, Time, Score, Level, Plate count (k/N), Gate state, and status lines for gems, boost, dash cooldown, checkpoint state. Temporary messages appear (e.g., “+25 Gem” or “Checkpoint reached!”).
7. **Camera Modes:** Orbit camera (arrow keys adjust angle/height). Follow camera (right mouse toggles; trails player from behind/above).
8. **Rotating Sweepers:** long red bars spinning around a pivot. If touched, you lose 1 life and respawn.
9. **Roaming Hazards (Movable Bars):** bars that move with a heading and speed, jitter direction randomly, and bounce off arena borders. May spin slowly. Collision checks distance-to-segment.
10. **Fragile Ground Tiles:** disks with states: intact → cracked → broken. Stand too long or step again soon = break. Broken acts as a hole. Falling costs 1 life and respawns you.
11. **Gems (Collectibles):** glowing yellow disks. Each gives +25 score. Collect all gems in a level for +1 extra life.
12. **Speed Pads (Boost):** cyan circles on the ground. Standing on one grants a speed boost (×1.8) for a few seconds. HUD shows boost timer.

13. **Checkpoint Pad:** large amber circle before the wall. Touch it to activate (turns green). After that, deaths respawn you here instead of the start.
14. **Teleports (Hole Rim Pads):** every hole gets a nearby teleport pad. Pads link in sequence: 1↔2, 3↔4, etc. If odd, last links back. Also adds a special wall-side pad linking to the first hole. Short cooldown avoids ping-ponging.
15. **Dash Ability:** press **Space** to burst forward in steps (ignores short obstacles if path clear). Cooldown (~1.2s) shown in HUD.
16. **Level System:** multiple levels with different layouts. Level 1 = 7 lives; Level 2 = 5 lives, shorter gate grace, faster hazards, and extra sweeper/roamer/hole.
17. **Game Over:** if Lives = 0 or Time = 0, screen shows **GAME OVER** with score. Press R to restart current level; number keys to jump to a level.
18. **Level Clear Menu:** after clearing, screen shows completion text. Options: **L** for next level, **R** to restart, **1** to return to Level 1. Level 2 shows a “Next level under implementation” message if chosen.
19. **Scoring Summary:**
- +25 per gem
 - +30 per plate (once)
 - +100 when gates open
 - +200 reaching gate
 - +1 life for all gems
 - +time bonus on clear